Switch Adapted Ms. Rachel Speak & Sing MAKER GUIDE

Makers Making Change
A Neil Squire Program

Required Components

1.	2.	3.	BOM
			1. Ms. Rachel Speak and Sing Toy (Amazon, Walmart) 2. 3.5mm Female Mono Cable (x2) 3. Cable Tie (x2) Refer to the Tool and Component List for purchasing details.

Required Tools

- Wire strippers
- Soldering iron and solder
- Flush Cutters
- Electrical Tape
- Seam ripper or scissors

Required Personal Protective Equipment (PPE)

- Safety glasses

BY-SA © 2025-2026 by Makers Making Change
This work is licensed under the CC BY SA 4.0 License: http://creativecommons.org/licenses/by-sa/4.0
Files available at https://github.com/makersmakingchange/Switch_Adapted_Toys

Switch Adapted Ms. Rachel Speak & Sing MAKER GUIDE

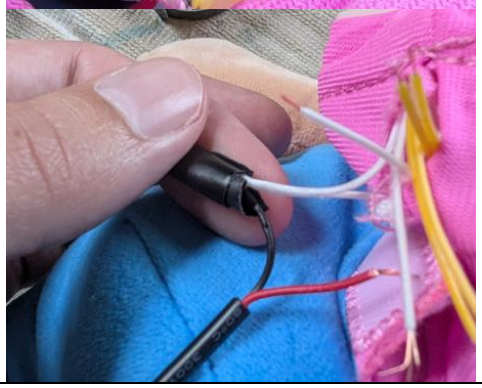
Assembly Instructions

1. Carefully remove the toy from its packaging without damaging it. Open the hook and loop fastener and pull the plastic box out of the back of the toy. Turn the slide switch to the “off” position while adapting the toy.	
2. Using the seam ripper, or a pair of scissors, open a small hole in the fabric below the plastic box, and pull the two wires that lead into the box out of the hole	
3. Using the flush cutters or wire strippers, cut both wires in half, about 4 cm from the plastic box.	

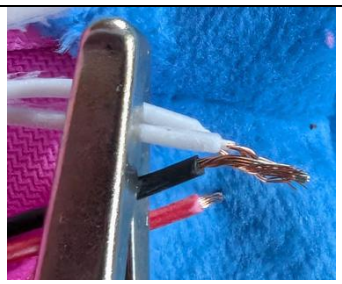
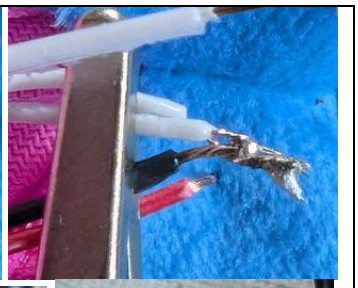
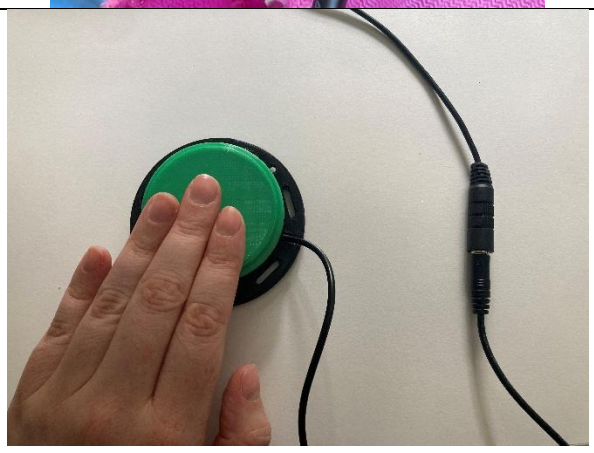
© 2025-2026 by Makers Making Change
This work is licensed under the CC BY SA 4.0 License: http://creativecommons.org/licenses/by-sa/4.0
Files available at https://github.com/makersmakingchange/Switch_Adapted_Toys

4. Using wire strippers, strip both ends of each of the cut wires, and separate the two strands in each wire half an inch down the wire.	
5. Take the mono cables, using the wire strippers, strip approximately 2 cm of the outer wire casing, revealing the internal wires. Skip this if the two internal wires are already showing.	
6. <u>If there are 2 internal wires</u>, strip approximately 0.5 cm of each wire. Skip this step if the internal wires have already been stripped, and the metal is exposed.	
7. <u>If there are three internal wires</u> (red, black, and exposed), strip off 1.5 cm of the red insulation and twist the red wire and exposed wire together. Strip 0.5 cm from the black wire.	 
8. Melt a small amount of solder over the exposed ends of the wire.	

© 2025-2026 by Makers Making Change
This work is licensed under the CC BY SA 4.0 License: http://creativecommons.org/licenses/by-sa/4.0
Files available at https://github.com/makersmakingchange/Switch_Adapted_Toys

9. Take a wire that leads to the box, and a wire **of the same colour** that goes into the toy. Twist these two wires together with one of the wires from the mono cable, then solder the three wires together. Give the wires a gentle pull to ensure they are fully soldered together and will not fall apart.	 
10. Take a one-inch piece of electrical tape, and wrap it around the solder joint to insulate it. Pinch the wrapped piece of tape to help squish it in place.	 

© 2025-2026 by Makers Making Change
This work is licensed under the CC BY SA 4.0 License: http://creativecommons.org/licenses/by-sa/4.0
Files available at https://github.com/makersmakingchange/Switch_Adapted_Toys

11. Repeat steps 9-10 with the other two wires of the same colour, and the second wire from the first mono cable. Take a wire that leads to the box, and a wire **of the same colour** that goes into the toy. Twist these two wires together with the second wire from the mono cable, then solder the three wires together. Take a one-inch piece of electrical tape, and wrap it around the solder joint to insulate it. Pinch the wrapped piece of tape to help keep it in place.	   
12. Repeat steps 9-11 with the other mono cable and the other colour wires. Each mono cable should connect only to one colour of wire. For each cable, the black wire should connect to each side of a cut wire, and the red to each side of the other cut wire of that colour.	
13. It is now time to test the toy. Slide the power switch to the “on” position and insert an assistive switch into one of the mono cables. The toy should activate when the assistive switch is pressed. Then, test the second mono cable. The mono cables replicate the buttons in the hands, where one causes it to speak and one causes it to sing.	

© 2025-2026 by Makers Making Change
This work is licensed under the CC BY SA 4.0 License: http://creativecommons.org/licenses/by-sa/4.0
Files available at https://github.com/makersmakingchange/Switch_Adapted_Toys

Troubleshooting notes:

- If the toy does not turn on, please check the soldered connections.
- If the toy continuously repeats the song when the switch is released, the connection is shorted. Check your connections and ensure the electrical tape is secure around each connection

14. Take the cable ties and tie each mono cable to the wires it is attached to. **Pull each cable tie as tight as possible** and use the flush cutters to trim off the ends.

    The cable ties are being used as strain relief, so that if the person playing with the toy pulls on the mono cable it will not pull apart the soldered joints, so it is important that the cable tie is tight or it will not protect the solder joint as expected.





15. Tuck the original cables and the spliced joints back through the hole in the fabric.



© 2025-2026 by Makers Making Change
This work is licensed under the CC BY SA 4.0 License: http://creativecommons.org/licenses/by-sa/4.0
Files available at https://github.com/makersmakingchange/Switch_Adapted_Toys

16. Tuck the plastic box back into the toy, and refasten the back, tucking the mono cables out the bottom of the hole	 
17. Test the toy again. If it is still working properly, repackage the toy and the adapted toy is complete!	

BY-SA © 2025-2026 by Makers Making Change
This work is licensed under the CC BY SA 4.0 License: http://creativecommons.org/licenses/by-sa/4.0
Files available at https://github.com/makersmakingchange/Switch_Adapted_Toys